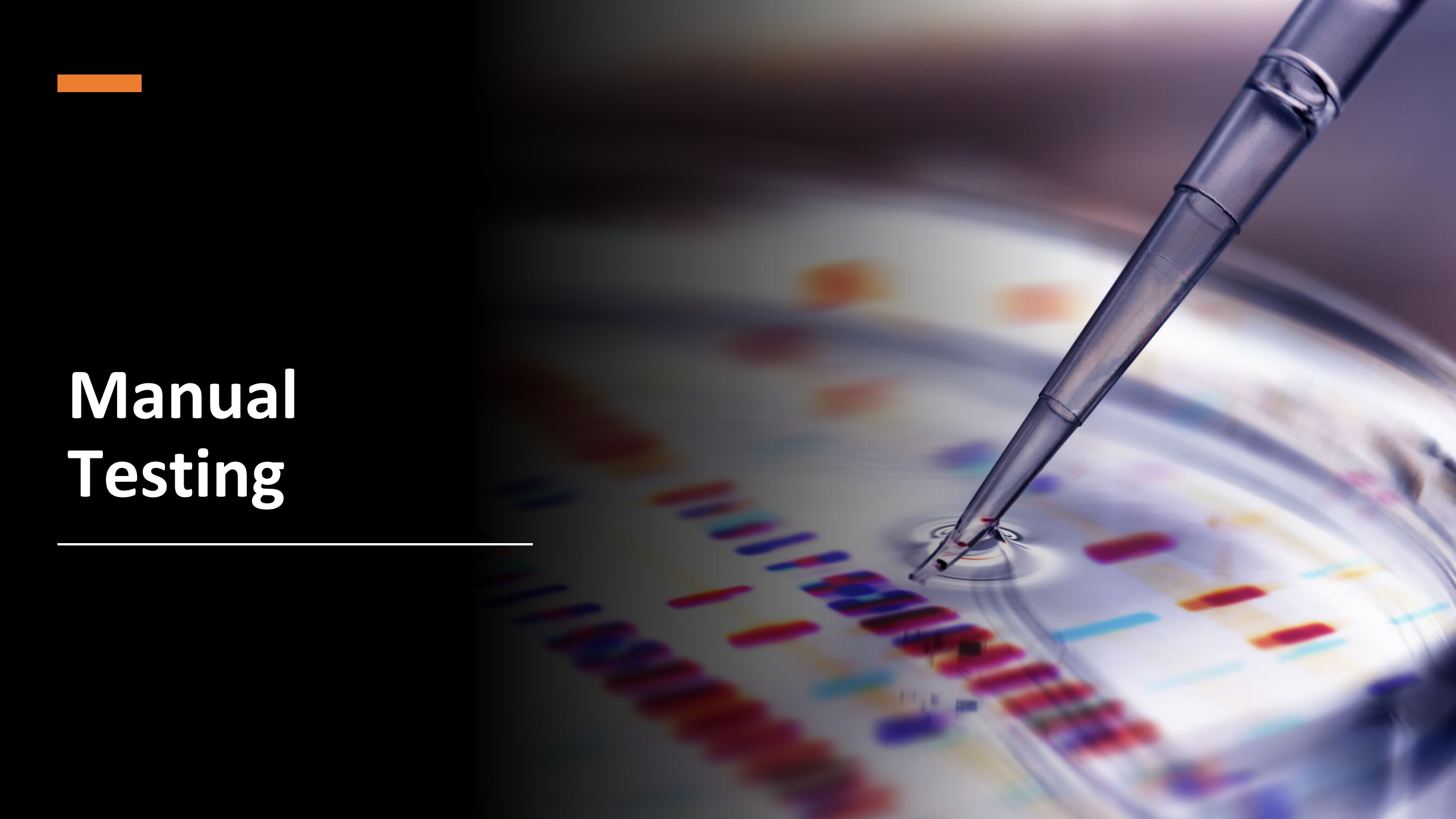
Manual Testing



Introduction

- Manual testing is a software testing process in which test cases are executed manually without using any automated tool. All test cases executed by the tester manually according to the end user's perspective. It ensures whether the application is working, as mentioned in the requirement document or not. Test cases are planned and implemented to complete almost 100 percent of the software application. Test case reports are also generated manually.
- Manual Testing is one of the most fundamental testing processes as it can find both visible and hidden defects of the software. The difference between expected output and output, given by the software, is defined as a defect. The developer fixed the defects and handed it to the tester for retesting.
- Manual testing is mandatory for every newly developed software before automated testing. This testing requires great efforts and time, but it gives the surety of bug-free software. Manual Testing requires knowledge of manual testing techniques but not of any automated testing tool.
- Manual testing is essential because one of the software testing fundamentals is "100% automation is not possible."

Need of Manual Testing Approach

- Whenever an application comes into the market, and it is unstable or having a bug or issues or creating a problem while end-users are using it.
- If we don't want to face these kinds of problems, we need to perform one round of testing to make the application bug free and stable and deliver a quality product to the client, because if the application is bug free, the end-user will use the application more conveniently.
- If the test engineer does manual testing, he/she can test the application as an end-user perspective and get more familiar with the product, which helps them to write the correct test cases of the application and give the quick feedback of the application.

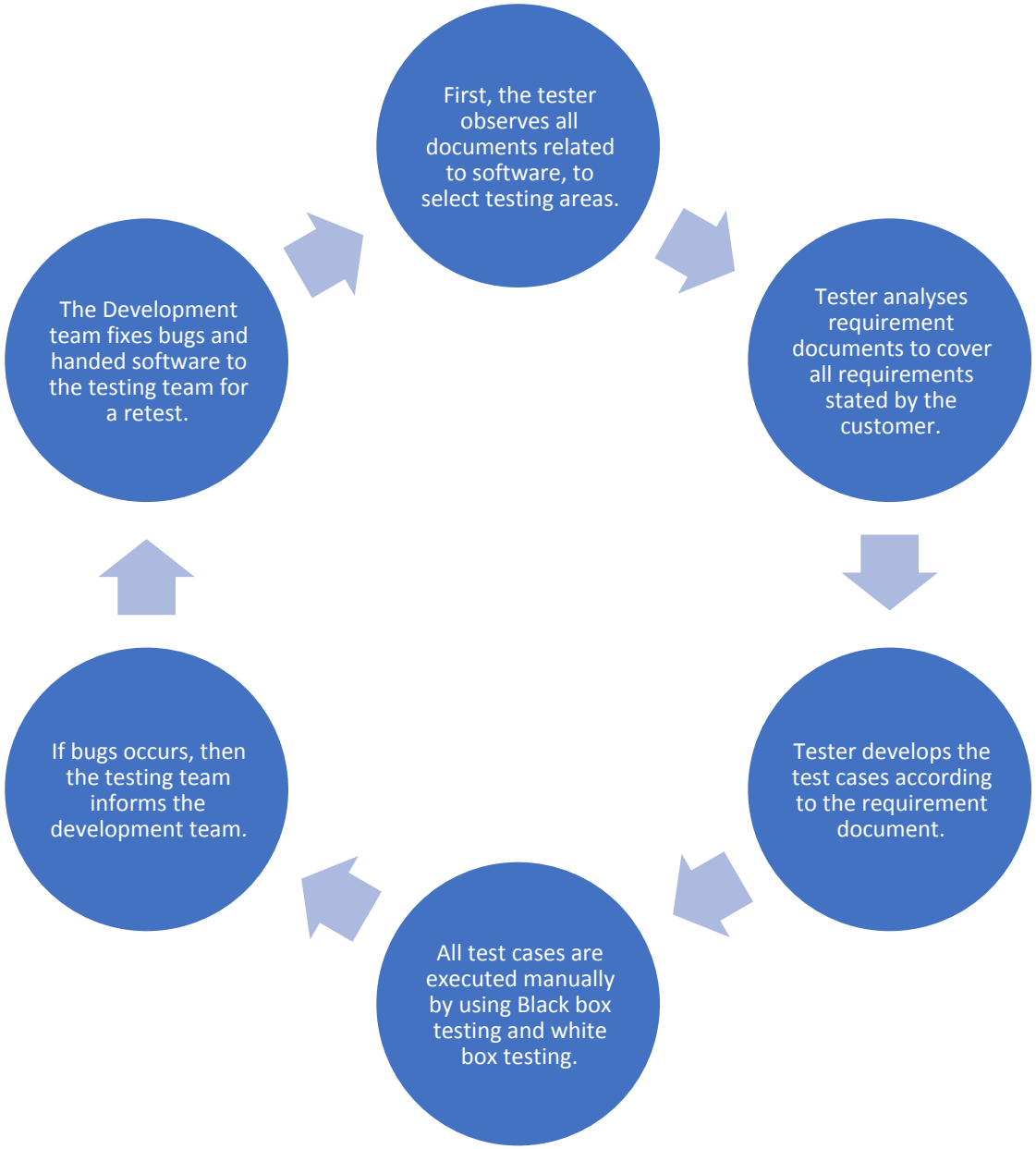
Types of Manual Testing

White Box Testing: The white box testing is done by Developer, where they check every line of a code before giving it to the Test Engineer. Since the code is visible for the Developer during the testing, that's why it is also known as White box testing.

Black Box Testing: The black box testing is done by the Test Engineer, where they can check the functionality of an application or the software according to the customer /client's needs. In this, the code is not visible while performing the testing; that's why it is known as black-box testing.

Gray Box Testing: Gray box testing is a combination of white box and Black box testing. It can be performed by a person who knew both coding and testing. And if the single person performs white box, as well as black-box testing for the application, is known as Gray box testing.

Steps to perform Manual Testing



Advantages of Manual Testing Approach

- It does not require programming knowledge while using the Black box method.
- It is used to test dynamically changing GUI designs.
- Tester interacts with software as a real user so that they have the ablility to discover usability and user interface issues.
- It ensures that the software is a hundred percent bug-free.
- It is cost-effective.
- Easy to learn for new testers.

Disadvantages of Manual Testing Approach

- It requires a huge number of human resources.
- It is very time-consuming.
- Tester develops test cases based on their skills and experience. There is no evidence that they have covered all functions or not.
- Test cases cannot be used again. Need to develop separate test cases for each new software.
- It does not provide testing on all aspects of testing.
- Since two teams work together, sometimes it is difficult to understand each other's motives, it can mislead the process.



Manual Testing Tools

- In manual testing, different types of testing like unit, integration, security, performance, and bug tracking, we have various tools such as Jira, Bugzilla, Mantis, Zap, NUnit, Tessy, LoadRunner, Citrus, SonarQube, etc. available in the market. Some of the tools are open-source, and some are commercial.
- In today's task as we are going use the testing platform "**JMeter**" so we would be discussing it here. For further information of the rest of the testing platforms please refer this link: https://www.javatpoint.com/software-testing-tools


LoadRunner
Citrus
Tessy
ZAP
Mantis
Manual
Testing Tools
NUnit
BugZilla
Jira
APACHE
JMeter
JMeter
SonarQube

Software Build Process

1. Once the requirement is collected, it will provide to the two different team development and testing team.
2. After getting the requirement, the concerned developer will start writing the code.
3. And in the meantime, the test engineer understands the requirement and prepares the required documents, up to now the developer may complete the code and store in the Control Version tool.
4. After that, the code changes in the UI, and these changes handle by one separate team, which is known as the build team.
5. This build team will take the code and start compile and compress the code with the help of a build tool. Once we got some output, the output goes in the zip file, which is known as Build (application or software).Each Build will have some unique number like (B001, B002).
6. Then this particular Build will be installed in the test server. After that, the test engineer will access this test server with the help of the Test URL and start testing the application.
7. If the test engineer found any bug, he/she will be reported to the concerned developer.
8. Then the developer will reproduce the bug in the test server and fix the bug and again store the code in the Control version tool, and it will install the new updated file and remove the old file; this process is continued until we get the stable Build.
9. Once we got the stable Build, it will be handed over to the customer.

Features of JMeter Software

- JMeter is an open-source tool that is used to test the performance of both static and dynamic resources and dynamic web applications.
  It is completely designed on the JAVA application to load the functional test behavior and measure the application's performance.
  It facilitates users or developers to use the source code for the development of other applications.
- It is platform-independent, which accepts a JVM like Windows, Mac, and Linux, etc.
  It supports a user-friendly GUI, which is interactive and straightforward.
  It is incredibly extensible to load the performance test in multiple types of servers.
  For more information about Jmeter, refers to the below link: https://www.javatpoint.com/jmeter-tutorial.



Steps to use JMeter Software

- Setting up Apache Jmeter. The first step is to go to the Apache Jmeter folder. ... Adding a Thread Group. Right-click on the Performance Testing. ... Adding HTTP Request. ... Adding Listeners. ... Running the Test.
- For more detailed information, please visit: https://www.google.com/url?sa=t&source=web&rct=j&url=https://www.simplilearn.com/tutorials/jmeter-tutorial/jmeter-performance-testing&ved=2ahUKEwjliMDKxeT6AhVgcGwGHXkJBnYQFnoECAgQBQ&usg=AOvVaw3MgsZlUEAVchbbhvoekdPt

Task to perform

- Explore the manual testing approach on “JMeter” platform and attempt to implement a manual testing approach on your on-going subject project and submit the solution in the form of pdf extension file in the Google Classroom.
- Testing Platform Link: https://www.google.com/url?sa=t&source=web&rct=j&url=https://jmeter.apache.org/&ved=2ahUKEwjQ7JG7w-T6AhXSSGwGHbw4Ah8QFnoECAoQAQ&usg=AOvVaw1kRUq-OnlWttcu1w7FWAGXension

